Blessed G/Michael Catholic Secondary School
1st Semester Biology Test Two for Grade 12th
Dec, 2018 E.C

Name Hanna Ayalew Section _____ No _____ Time Allowed 1 hour

I. Choose the best answer among the given alternatives.

D 1. Which of the following statements concerning archaens is true of the new classification of living things into three domains of life?
A .resembles eukaryotes in all aspects ✗
B .completely differs from prokaryotes ✗
C .completely differ from both eukarya and bacteria ✗
D .have some novel features that are absent in eukarya and bacteria ✓

B 2. If the experimental population of E coli lacks F – factor (F-plasmid) ,and if bacteriophages are excluded from the bacterial cultures ,then beneficial mutations might be transmitted horizontally to other E coli cells via ----
A .conjugation ✗ B .transformation C .sex pili ✗ D .transduction

C 3. Which of the following does not occur during binary fission in bacteria?
A .cytokinesis ✓ B .cell elongation ✓ C .spindle formation ✗ D .DNA replication ✓

A 4. Which of the following microorganisms thrive under extreme environmental conditions like high temperature, acidic PH, lack of oxygen and high salt concentration?
A .archae bacteria B .cyanobacteria C .eubacteria D .all

A 5. The cocci which forms a chain is?
A .strepto cocci B .staphylococci C.diplococci D . Tetra cocci

A 6. Gram positive bacteria have ;
A .multiple layer of peptidoglycan that helps to retain the crystal violet stain ✓
B .second outer membrane that helps to retain the crystal violet stain ✗
C .thick capsule that traps the crystal violet stain ✗
D .periplasmic space that traps the crystal violet stain ✗

C 7. Which of the following does not contain a protein?
A .pili ✗ B .flagellum ✗ C . lipotechoic acid D .porins ✗

C 8. Which method of horizontal gene transfer involves a virus acting as a vector?
A .transformation ✗ B .conjugation ✗ C .transduction ✓ D .binary fission ✗

C 9. What is the main function of the decolrizing agent (alcohol) in gram staining?
A .to fix crystal violet iodine complex to the cell wall ✗
B .to wash away the safranin stain from gram negative bacteria ✗
C .to wash away the crystal violet stain from gram negative bacteria ✓
D .to wash away the crystal violet stain from gram positive bacteria ✗

A 10. Gram negative bacteria are more resistant to antibiotics due to the presence of ?
A .outer polysaccharide layer ✓ C .porin proteins ✗
B .thin peptidoglycan cell wall ✗ D .techoic acid ✗

C 11. Assuming that the microorganism possess a cell wall on which prokaryotes should be expected and most strongly resistant to plasmolysis in hypertonic environment the bacteria is most probably?
A .extreme thermophile B .extreme halophiles C .cyanobacteria D .methanogens

A 12. Plant like photosynthesis that releases oxygen occurs in?
A .cyanobacteria ✓ B .archae ✗ C .chemoautotrophic bacteria ✗ D .all

A 13. Which of the following best describes an organism that obtains its energy from the oxidation of inorganic compounds and uses CO_2 as its sole carbon source?
A .chemolithoautotrophs C .chemolithohetotophs
B .photoorganohetrotophs D .chemoorganoheytrotrophs

A 14. Which nutritional type would be most likely to be found in the deep sea hydrothermal vents, where sunlight is absent and chemical compounds like H_2S are abundant?
A .chemolithoauthotophs C .photolithoautotophs
B .chemoorganohetrotrophs D .photoorganotrophs

B 15. Endospores are all of the following except as compared to vegetative cells?
A .more likely to survive treatment with disinfectants ✓
B .more likely to die in nutritionally poor conditions ✗
C .more resistant to staining ✓
D .more resistant to temperature changes ✓

C 16. Which of the following contains DNA?
A .Gm- cell wall ✗ B .pili ✗ C .plasmids ✓ D .flagella ✗

D 17. The bacterial envelope includes all of the following except?
A .capsule ✓ B .cell wall ✓ C .cell membrane ✓ D .endospore ✗

B 18. The term micro biota refers to which of the following /
A .all of microorganisms involved in symbiotic relationship ✗
B .normal microbial flora that is found on skin, gut and mucous membrane of human
C.all microorganisms in a certain geographic area ✗
D .all microorganisms of the same species ✗

C 19. During which phase of viral replication does the virus obtain its envelope?
A .release B .attachment ✗ C .assembly ✓ D .penetration ✗

B 20. Which of the following is the genome of polio virus?
A .SsRNA ✓ B.dsDNA C.SsDNA D .dsRNA ✗

C 21. Which statement is true about a viroin?
A .it is a lytic phage ✗
B .it is a lysogenic phage ✗
C .it is an infective and fully formed viral particle ✓
D .it relies on the viral capsid ✗

A 22. An icosahedral capsid is made up of?
A .triangular capsomeres C .pentagonal capsomeres
B.hexagonal capsomeres D .B&C

B 23. What are infectious RNA particles without a protein coat called?
A .viroin B .viroid C .prions D .none

A 24. The lytic cycle of a bacteriophage is distinct from the lysogenic cycle because the lytic cycle involves /
A .synthesis of new phage particles followed by lysis of the host cell ✓
B .integration of the phage DNA into the host chromosome ✗
C .replication of the Phage DNA in synchronize with the host's cell ✗
D .the phage DNA remaining dormant as a prophage for several generations ✗

C 25. Temperate bacteriophages like lambda phages can exist as a prophage, in this stage the phage is characterised by?

A .active production of viroins ✗

B .immediate lysis of the host bacterium ✗

C .integration of its genome into the host bacterium genome ✓

D .having a double strand RNA genome ✗

C 26. The host range of a virus is primarily determined by?

A .type of nucleic acid it contains ✗

B .presence of a lipid envelope ✗

C .interaction between viral surface proteins and host cell receptors ✓

D .size of a virus ✗

B 27. What is the killing of aquatic life due to the depletion of oxygen because of the excessive growth of algal due to the overabundance of phosphorous in the coastal areas called?

A.leaching B.eutrophication C .respiration D .photosynthesis

D 28. How is carbon stored in the biosphere?

A .in the atmosphere as carbon dioxide ✓ C. Underground as fossils ✓

B .in oceans as dissolved carbon dioxide ✓ D .all

A 29. Nitrates in the soil can be absorbed by plants; nitrates can also be removed from the soil by the action of?

A .denitrifying bacteria in poorly aerated conditions ✓

B .bacteria in the root nodules of legumes ✓

C .nitrogen fixing bacteria on ammonium compounds ✗

D .nitrogen fixing bacteria on decaying plants ✗

A 30. Which of the following is the simplified sequence of phosphorous cycling in a terrestrial ecosystem?

A .rock mineral --→soil →producers →consumers →decomposers

B. soil →producers →rock minerals →consumers →decomposers

C .decomposers→ rock minerals →consumers' →producers' D .none

D 31. Which of the following protozoan causes African sleeping sickness?

A .Plasmodium falciparum C .Trypanasoma cruzei

B .Trichomonas vaginalis D . Trypanasoma brucei

B 32. Which of the following parasite is transmitted to humans through exposure to cat faeces?

A .Trichomonas vaginalis C .Endamoeba histolytica

B .Toxoplasma gondi D .plasmodium falciparum

C 33. Asexual spores of fungi with in a sac are called?

A sporangiospores B .conidiospores C .arthro spores D .all

D 34. Rhizopus , penecilum and asperigilus are examples of

A .symbiotic fungi C . Saprophytic fungi

B .parasitic fungi D .all

B 35. Which of these spores are characteristics of the black bread mold rhizopus ?

A .basidospores ✗ B .ascospores C .zygospores D .oospores ✗

B 36. Thick walled ,single celled spores which are highly resistant to adverse conditions is /

A .zoospores B .chlamydospores C .arthro spores D .basido spores

A 37. Ring worms of in the hairless in our body is called?

A .Tinea corporis B .Tinea captitis C .Tinea barbae D .Tinea ungium

38/ Which of the following statement concerning Candida albicans is false /

A .C. albicans is budding yeast that forms pseudo hyphae when it invades tissue

B .C, albicans causes thrush

C .C.albicans is transmitted primarily by respiratory aerosol

D .C .albicans causes opportunistic infections

39/ Tinea captitis may be caused by all the following except?

A .Microsporum gypseum C .Trichophyton mentagrophytes

B .Trichophyton verruccosum D .Epidermophyton floccosum

A 40. Which of the following is non hyphal fungus?

A .yeast B .mucor C .rhizopus D .mushroom

B 41. Which of the following Achaeans are obligate anaerobes?

A .halophiles B .methanogens C .thermophiles D .alkaliphiles

A 42. Which method achieves complete sterilization?

A .tyndallisation C .pasteurization

B .autoclaving at 121degree Celsius for 20 minutes D .using alcohol

A 43. Bacterial endospores are more resistant to control methods because?

A . they are metabolically in active with protective coats

B .They have a waxy cell wall

C They contain multiple cell membranes D .none

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Blessed G /Michael Catholic secondary BIOLOGY MID 2 ANSWER SHEET FOR GRADE 12

NAME ---------- SECTION ---------- ROLL NUMBER ----

NO	ANSWER	NO	ANSWER	NO	ANSWER	NO	ANSWER	NO	ANSWER
1	D	11	C B	21	C	31	D	41	B
2	B	12	A	22	A D	32	B	42	A B
3	C	13	A	23	B	33	C	43	A
4	A	14	A	24	A	34	D C		
5	A	15	B	25	C	35	B C		
6	A	16	C	26	C	36	B		
7	C	17	D	27	B	37	A		
8	C	18	B	28	D	38	D C		
9	C	19	C	29	A	39	D		
10	A	20	B A	30	A	40	A		